FM-7 / FM-NEW7
Schematics Companion

English translation index for the original Japanese sheets, plus a motherboard IC reference

Source	**I/O (アイ・オー) magazine, April 1983** (1983年04月号), pp. 281-300
Article	FM-7 全回路図 — *FM-7 Complete Circuit Diagrams*
Compiled PDF	`Documents/IO_198304_FM-7_Schematics.pdf` — 20 pages; PDF p.1 = magazine p.281
Board	Fujitsu FM-NEW7 motherboard, 85 IC positions (M1-M85)

Why this document exists

The schematics themselves are drawn almost entirely in Latin script — part numbers, pin names and net names are already in English, so they can be read directly by a non-Japanese speaker. **The only Japanese on the sheets is the title printed at the top of each page**, plus the contents list and the editor's note on the title page. This document translates all of it, indexes the twenty sheets, and adds the board's IC list so that a reference designator seen on a sheet can be looked up without leaving the bench.

Editor's note, title page (p.281)

> 本資料は編集部の責任において、あくまでも参考資料として掲載しました。したがって、本資料に関してメーカーに直接問い合わせることはご遠慮ください。（編）
>
> *“This material is published on the editorial department's own responsibility, purely as reference material. Please therefore refrain from contacting the manufacturer directly with enquiries about it. — Ed.”*

Running header on every page

Japanese	English
FM-7 回路図	FM-7 Circuit Diagrams
FM-7 全回路図	FM-7 Complete Circuit Diagrams (title page)
目次	Contents

A caution on using FM-7 schematics with an FM-NEW7 board

These drawings are for the **original FM-7**. The FM-NEW7 is functionally the same machine but was cost-reduced using higher-integration LSI and gate arrays, so discrete logic on these sheets has in places been absorbed into custom silicon (positions M38, M49, M55, M72, M79 and M80 in the IC list are all undocumented Fujitsu gate arrays). Net names, signal polarity, bus structure and the memory/I/O map transfer well. **Reference designators and chip-level placement do not** — the M-numbers on these sheets are FM-7 positions, not FM-NEW7 ones.

Contents of this document

1. Quick sheet index — all twenty sheets on one page
2. Sheet-by-sheet detail — title translation and what is drawn on each sheet
3. Reading the sheets — naming conventions and symbols
4. Boot mode and clock DIP switches — FM-7 and FM-NEW7
5. Fault-finding shortcuts — which sheet answers which question
6. FM-NEW7 motherboard IC list — 85 positions, M1 to M85
7. Device summary — counts and memory totals

1. Quick sheet index

PDF page numbers refer to the compiled 20-page scan; magazine pages are the original I/O pagination. The offset is constant: **PDF page = magazine page − 280**.

PDF	Mag.	Japanese sheet title	English
1	p.281	FM-7 全回路図 / 目次	Title page & contents
2	p.282	メインCPU部	Main CPU section
3	p.283	クロック制御部	Clock control section
4	p.284	メインI/Oアドレス部	Main I/O address (decode) section
5	p.285	メインROM部	Main ROM section
6	p.286	メインRAM部	Main RAM section
7	p.287	メイン・サブ部	Main-Sub interface section
8	p.288	メイン プリンタ・カセット部	Main printer & cassette section
9	p.289	共有メモリ部 / サブCPU部	Shared-memory section / Sub-CPU section
10	p.290	サブ アドレス・デコード部	Sub address decode section
11	p.291	サブROM/RAM部	Sub ROM / RAM section
12	p.292	サブCRT CNTRL部	Sub CRT controller section
13	p.293	サブREG/FLAG部 / サブCRTアドレス部	Sub register/flag section / Sub CRT address section
14	p.294	サブCRT RAM部	Sub CRT RAM (VRAM) section
15	p.295	サブCRTインターフェイス部	Sub CRT interface section
16	p.296	キーインターフェイス部	Keyboard interface section
17	p.297	コネクタ(拡張Z80)部 / コネクタPSG部	Connector section (Z80 expansion) / PSG connector section
18	p.298	コネクタ(RS-232C, 漢字)部 / 拡張バス・バッファ部	Connector section (RS-232C, Kanji ROM) / Expansion-bus buffer section
19	p.299	メインPSG部	Main PSG (sound) section
20	p.300	電源部	Power-supply section

2. Sheet-by-sheet detail

Sheet	Title	What is drawn on the sheet
PDF 1 p.281	FM-7 全回路図 / 目次 **Title page & contents**	Photo of the system, editor's note, contents list.
PDF 2 p.282	メインCPU部 **Main CPU section**	MB68B09 main CPU, address/data bus buffers (MB74LS244/245), interrupt & control pull-ups (RB8), power-on reset (D1 1S4468, C5, MB74LS14), MCPUCLK to EXTAL, Z80 / *DMA / *MRDY / *HALT handling.
PDF 3 p.283	クロック制御部 **Clock control section**	X1 4.9152 MHz oscillator, SN74LS93 divider (2 MHz / 1.2 MHz taps), SW2 speed select (SCLK1/SCLK2), generation of MCPUCLK, SCPUCLK, *RESETB, EB, 2MS, *MRDY, MB84040B counter, interrupt-mask/timer logic.
PDF 4 p.284	メインI/Oアドレス部 **Main I/O address (decode) section**	MB74LS138/LS27/LS30 address decoders producing *FD00-*FD07, *RFD00-*RFD05, *WFD00-*WFD03, *IOS, *WTQE, *RDE, *PL_TREG.
PDF 5 p.285	メインROM部 **Main ROM section**	MB83256 (32 K) F-BASIC ROM, MB7063/MB8561 boot ROMs, SW1 boot-mode select, *BTROM, *SUBSEL, *MOS, *RAMHB0/*RAMHB1.
PDF 6 p.286	メインRAM部 **Main RAM section**	8 x MB8265A-15 (64 K x 1 DRAM) = 64 KB main RAM, row/column address mux (MB74LS158), *RAS/*CAS/*WE generation, refresh (*REFCNT, *REFCK).
PDF 7 p.287	メイン・サブ部 **Main-Sub interface section**	MB14415 main-side interface gate array, *SCLKNMI, *DMA, *BREAK, BUSY, *ATTENT, *EXTDET, *RESETB, buzzer/sound gating, *GRNT/Z8W handshake between the two CPUs.
PDF 8 p.288	メイン プリンタ・カセット部 **Main printer & cassette section**	Centronics printer port (CN5, SN74LS273 latches, LPBUSY, *LPNT, *PGCLR), cassette interface (CN6 DIN, MB3614 op-amp, *CANCEL, *SUBHALTREQ), *RESETB.
PDF 9 p.289	共有メモリ部 / サブCPU部 **Shared-memory section / Sub-CPU section**	**Upper:** the 128-byte shared window (MB74LS244/245 buffers, *SHALTAC, *SSMEM, *SUBSEL, *SRDQE/*SQANDE) between main and sub address/data buses. **Lower:** MC68B09 sub CPU, SCPUCLK to EXTAL, *SRESET (from *RESETB via MB74LS08), *SHALT, *SCLKNMI, *SUBIRQ, *KSTROBE, sub bus buffers.
PDF 10 p.290	サブ アドレス・デコード部 **Sub address decode section**	MB74LS138 decoders for the sub CPU map: *SROMSEL, *SRAMCS, *SRAMOE, *SSMEM, VRAM *SDRAM(R/G/B), *SCRTSW, *SVRACS, *SLED, *SBUSYSET, *KACKING, *BUZZER, DL1 delay line for SCPURAS/SCPUCAS.
PDF 11 p.291	サブROM/RAM部 **Sub ROM / RAM section**	MBM2764-20 and MBM2732A-20 sub-monitor ROMs (J1/J2 jumpers select *SROMSEL), 2 x MB8128 (2 K x 8) sub SRAM.
PDF 12 p.292	サブCRT CNTRL部 **Sub CRT controller section**	MB61030 (MB610301) CRTC gate array, X2 16.128 MHz dot clock, SVRAS/SVCAS, *SCSYNC, *SBLANK, SCLK1-SCLK3, *SVSYNC/*SHSYNC, *SVDHALT, *SFTCLK, *SLOAD, register select (*SREGL/*SREGH, SADRSEL).
PDF 13 p.293	サブREG/FLAG部 / サブCRTアドレス部 **Sub register/flag section / Sub CRT address section**	**Upper:** MB74LS74A flip-flops for *SVDOFF, SRWB, SVOHALT, *SHALTST/SHALTAC, BUSY, *SUBIRQ, *CANSEL, *SIRQCLR, and the SN74LS273 page/plane register (*VPAGE1-3, DPAGE1-3). **Lower:** MB8462 VRAM address mux, plane selects *SVCASB/R/G, DL2 delay line, J9/J10 jumpers, SADRSEL.
PDF 14 p.294	サブCRT RAM部 **Sub CRT RAM (VRAM) section**	The 48 KB video RAM: three 16-device banks of MB81416 (16 K x 4 DRAM) - blue, red and green planes - with MB74LS244 data buffers and per-plane *SDRAM/*SVCAS strobes.
PDF 15 p.295	サブCRTインターフェイス部 **Sub CRT interface section**	MB13501 video output gate array, MB74LS166 shift registers (dot serialisation), RGB drive transistors (2SC1213), CN4 digital RGB connector, CN6 8-pin DIN, sync mixing (*SHSYNC, *SVSYNC, SVIDEOCLK).
PDF 16 p.296	キーインターフェイス部 **Keyboard interface section**	MB88401 keyboard-scan microcontroller, CN6 keyboard connector, MB74LS145 row drivers, *KSTROBE, *KEYIN, *KACKING, *KDATA9, *BREAK, *LPMASK/*TIMMASK, SCLK2.
PDF 17 p.297	コネクタ(拡張Z80)部 / コネクタPSG部 **Connector section (Z80 expansion) / PSG connector section**	**Upper:** CN7 50-pin Z80-card slot and CN8/CN9 (EADDRBUS, EDATABUS, *EIRQ, *ENMI, *EFIRQ, EBA/EBS, ERW, *EMRDY, EQ/EE, Z80↓, *EEXTDET). **Lower:** CN8A/CN8B PSG board connector (AUDOUT, 1.2MHz, *RESETB, EB).
PDF 18 p.298	コネクタ(RS-232C, 漢字)部 / 拡張バス・バッファ部 **Connector section (RS-232C, Kanji ROM) / Expansion-bus buffer section**	**Upper:** CN12/CN13 A- and B-side option connectors (EAB0-7, EE, *ERESET, *EEXTDET, ~2.3 MHz, ±12 V, +5 V). **Lower:** MB74LS245/LS244/LS241 buffers isolating MADDRBUS/MDATABUS from EADDRBUS/EDATABUS, plus interrupt pass-through (*EIRQ, *EFIRQ, *ENMI, *EMRDY).
PDF 19 p.299	メインPSG部 **Main PSG (sound) section**	AY-3-8910 PSG, LM386 audio amp, CN1/CN4 audio out, MB74LS74/LS244 register and address latches, BC1/BDIR control, *WFD0E/*RFD0E/*WFD0D, 1.2MHz clock, *RESETB.
PDF 20 p.300	電源部 **Power-supply section**	CN10 PSU input (+12 V, +5 V, -12 V, GND), MC7805CT regulator, bulk caps C24/C25/C27/C28, and the decoupling-capacitor allocation blocks - each box lists which board positions (M1, M2, M5 ...) the 0.1 µF caps CB1, CB39, CB43, CB56 etc. belong to.

3. Reading the sheets

- **A leading asterisk means active low.** `*RESETB`, `*SHALT`, `*IRQ` and so on are all asserted low. The original sheets use an overbar; the typesetting renders it as a leading asterisk.
- `M...` **prefix means main-CPU side,** `S...` **means sub-CPU side,** `E...` **means expansion bus.** So `MADDRBUS`/`MDATABUS` versus `SADDRBUS`/`SDATABUS` versus `EADDRBUS`/`EDATABUS`.
- `M<number>` **is a board reference designator** (M1, M28, M75 ...), matching the silkscreen on the PCB. `RB<n>` is a resistor network, `CB<n>` a decoupling capacitor, `DL<n>` a delay line.
- **Double-arrow symbols at the sheet edges are off-sheet net connectors.** Follow the net name to the sheet that drives it; the quick index on the previous page is the fastest way to find it.
- **Fujitsu** `MB74LSxx` **parts are second-sourced 74LSxx logic.** Four-digit `MB....` parts (MB14415, MB61030, MB13501, MB8462, MB88401) are Fujitsu custom gate arrays with no public datasheet.

4. Boot mode and clock DIP switches

The two machines do not agree here, so both are given. On the FM-7 sheets the boot-mode switch appears on the Main ROM sheet and the speed switch on the Clock Control sheet; on the FM-NEW7 all four functions sit in a single 4-way SW2.

FM-7, as drawn on these sheets

Switch	Sheet	Function
SW1	PDF 5 / p.285	Boot-mode select — drives `*BTROM` and the ROM/RAM mapping of $8000-$FBFF
SW2	PDF 3 / p.283	CPU speed select — `SCLK1`/`SCLK2` taps off the SN74LS93 divider

FM-NEW7, as found on the board (SW2, 4-way)

Position	Function
SW2-1	Boot ROM A9 address (off = high, on = low)
SW2-2	Boot ROM A10 address (off = high, on = low)
SW2-3	Unknown purpose
SW2-4	Clock speed (off = 2 MHz, on = 1 MHz for FM-8 compatibility)

SW2-1 and SW2-2 ground address lines A9 and A10 of the 2 KB boot ROM, selecting one of four 512-byte boot blocks. Those same lines also preset the flip-flop that decides whether $8000-$FBFF presents BASIC ROM or RAM, which is why a boot-mode change alters behaviour before any ROM code runs.

5. Fault-finding shortcuts

Question	Sheet
Where do the main CPU's reset, HALT, MRDY and DMA come from?	**PDF 2** / p.282
Is the CPU clock reaching EXTAL, and how is `*RESETB` generated?	**PDF 3** / p.283
Why is the sub CPU halted; what drives `*SHALT` and `*SRESET`?	**PDF 9** / p.289, with the handshake on **PDF 7** / p.287
Which decoupling cap and rail belongs to a given IC position?	**PDF 20** / p.300
How is the BASIC ROM / RAM mapping of $8000-$FBFF switched?	**PDF 5** / p.285, decode on **PDF 4** / p.284
Where is the main DRAM refresh and RAS/CAS timing generated?	**PDF 6** / p.286

6. FM-NEW7 motherboard IC list

85 positions in silkscreen order. The *Equivalent* column gives the generic industry part for sourcing a replacement. Blank *Usage* cells are ordinary glue logic with no special role; entries marked *Unknown* are genuinely undocumented rather than omitted.

Pos.	Marking	Equivalent	Function	Usage / notes
M1	M74LS273P	74LS273	Octal D flip-flop with clear	
M2	MB74LS05	74LS05	Hex inverter, open-collector outputs	
M3	MB74LS245	74LS245	Octal bus transceiver, tri-state	
M4	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M5	MB74LS14	74LS14	Hex Schmitt-trigger inverter	
M6	MB74LS04	74LS04	Hex inverter	
M7	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M8	MB74LS14	74LS14	Hex Schmitt-trigger inverter	
M9	MB74LS14	74LS14	Hex Schmitt-trigger inverter	
M10	MB74LS367A	74LS367A	Hex bus driver, tri-state (4+2 enable)	
M11	SN7407N	7407	Hex buffer / driver, open-collector, high-voltage outputs	
M12	MB74LS74A	74LS74A	Dual D flip-flop with preset and clear	
M13	MB74LS02	74LS02	Quad 2-input NOR gate	
M14	386D	LM386 / NJM386D	Low-voltage audio power amplifier	Internal speaker output stage - amplifies beep and PSG audio
M15	SN7407N	7407	Hex buffer / driver, open-collector, high-voltage outputs	
M16	SN7407N	7407	Hex buffer / driver, open-collector, high-voltage outputs	
M17	MB3614	LM324	Quad operational amplifier, 14-pin	Analogue signal conditioning - cassette (CMT) read amplifier / comparator and audio mixing
M18	MB74LS08	74LS08	Quad 2-input AND gate	
M19	MB74LS175	74LS175	Quad D flip-flop with clear	
M20	MB74LS00	74LS00	Quad 2-input NAND gate	
M21	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M22	MB74LS74A	74LS74A	Dual D flip-flop with preset and clear	
M23	MB74LS175	74LS175	Quad D flip-flop with clear	
M24	MB74LS367A	74LS367A	Hex bus driver, tri-state (4+2 enable)	
M25	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M26	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M27	MB74LS374	74LS374	Octal D flip-flop, tri-state outputs	
M28	AY-3-8910	AY-3-8910	3-voice PSG sound generator with two 8-bit parallel I/O ports	Built-in sound, on the sub-CPU side; PSG I/O ports serve the joystick interface
M29	MB15021	—	Custom digital colour palette LSI	Maps the 8 colour codes (0-7) to RGB output
M30	MB74LS132	74LS132	Quad 2-input NAND Schmitt trigger	
M31	MB84049B	CD4049B	CMOS hex inverting buffer / level shifter	Unknown - conventionally level translation on the cassette or external interface
M32	MB434	SN75452 (?)	Dual peripheral / power driver, 8-pin, open-collector outputs	Ch.1 (pin 3) drives the buzzer; Ch.2 (pin 5) drives the cassette motor relay (??)
M33	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)

Pos.	Marking	Equivalent	Function	Usage / notes
M34	MB74LS240	74LS240	Octal buffer / line driver, tri-state, inverting	
M35	MB74LS138	74LS138	3-to-8 line decoder / demultiplexer	
M36	HD68B09P	MC68B09	8-bit CPU, 2MHz rated	Sub-CPU - display, VRAM, keyboard and PSG
M37	MB74LS245	74LS245	Octal bus transceiver, tri-state	
M38	MB112T605	—	Fujitsu semi-custom gate array	Unknown
M39	MB60H010A	—	Custom 64-pin LSI	Likely the display controller (the FM-7 has no standard 6845)
M40	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M41	MB8128-15	6116	2K x 8 bit SRAM, 150ns	Sub-system work RAM / inter-CPU shared RAM
M42	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M43	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M44	MB74LS04	74LS04	Hex inverter	
M45	MB74LS04	74LS04	Hex inverter	
M46	MB74LS138	74LS138	3-to-8 line decoder / demultiplexer	
M47	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M48	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M49	MB112T604	—	Fujitsu semi-custom gate array	Unknown
M50	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M51	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M52	MB8128-15	6116	2K x 8 bit SRAM, 150ns	Sub-system work RAM / inter-CPU shared RAM
M53	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M54	HD68B09P	MC68B09	8-bit CPU, 2MHz rated	Main CPU - runs the boot ROM and F-BASIC
M55	MB112T603	—	Fujitsu semi-custom gate array	Unknown
M56	MB74LS05	74LS05	Hex inverter, open-collector outputs	
M57	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M58	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M59	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M60	MB83256 (167)	32K x 8 mask ROM (27256 footprint)	32KByte mask ROM	Character generator & sub-system monitor
M61	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M62	MB14415	—	Unknown	Unknown
M63	MB74LS138	74LS138	3-to-8 line decoder / demultiplexer	
M64	MB74LS158	74LS158	Quad 2-to-1 multiplexer, inverting outputs	
M65	MB74LS158	74LS158	Quad 2-to-1 multiplexer, inverting outputs	
M66	MB74LS245	74LS245	Octal bus transceiver, tri-state	
M67	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M68	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M69	MB81416-12	4416	16K x 4 bit DRAM, 120ns	Graphics VRAM (6 devices = 48KB)
M70	MB74LS08	74LS08	Quad 2-input AND gate	
M71	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M72	MB111S601	—	Fujitsu semi-custom gate array	Unknown
M73	MB8516	2716	2K x 8 EPROM / mask ROM	Boot ROM (four 512 byte blocks)

Pos.	Marking	Equivalent	Function	Usage / notes
M74	MB83256 (160)	32K x 8 mask ROM (27256 footprint)	32KByte mask ROM	F-BASIC V3.02
M75	MB74LS30	74LS30	8-input NAND gate	
M76	MB88401 (709M)	MB8840-series 4-bit MCU	Single-chip 4-bit microcontroller with internal mask ROM	Keyboard controller - key matrix scanning, key-code generation and LED control
M77	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M78	MB8265A-15	4164	64K x 1 bit DRAM, 150ns	Main RAM (8 devices = 64KB)
M79	MB112T602	—	Fujitsu semi-custom gate array	Unknown
M80	MB112T601	—	Fujitsu semi-custom gate array	Unknown
M81	MB74LS245	74LS245	Octal bus transceiver, tri-state	
M82	MB74LS244	74LS244	Octal buffer / line driver, tri-state, non-inverting	
M83	MB8128-15	6116	2K x 8 bit SRAM, 150ns	Sub-system work RAM / inter-CPU shared RAM
M84	MB74LS145	74LS145	BCD-to-decimal decoder / driver, open-collector, high-voltage	
M85	MB74LS145	74LS145	BCD-to-decimal decoder / driver, open-collector, high-voltage	

7. Device summary

Memory complement

Device	Qty	Organisation	Total	Role
MB8265A-15	8	64K x 1 DRAM, 150ns	**64 KB**	Main RAM
MB81416-12	6	16K x 4 DRAM, 120ns	**48 KB**	Graphics VRAM
MB8128-15	3	2K x 8 SRAM, 150ns	**6 KB**	Sub-system work / shared RAM
MB83256	2	32K x 8 mask ROM	**64 KB**	F-BASIC V3.02; chargen + sub-monitor
MB8516	1	2K x 8 EPROM	**2 KB**	Boot ROM, 4 x 512-byte blocks

The 64 KB and 48 KB figures match the published specification exactly, which is a useful cross-check that no device has been mis-identified.

Logic device counts

Marking	Qty	Marking	Qty	Marking	Qty
MB74LS244	11	MB74LS175	2	MB3614	1
MB8265A-15	8	MB74LS367A	2	MB434	1
MB81416-12	6	MB74LS74A	2	MB60H010A	1
MB74LS245	4	386D	1	MB74LS00	1
MB74LS04	3	AY-3-8910	1	MB74LS02	1
MB74LS138	3	M74LS273P	1	MB74LS132	1
MB74LS14	3	MB111S601	1	MB74LS240	1
MB8128-15	3	MB112T601	1	MB74LS30	1
SN7407N	3	MB112T602	1	MB74LS374	1
HD68B09P	2	MB112T603	1	MB83256 (160)	1
MB74LS05	2	MB112T604	1	MB83256 (167)	1
MB74LS08	2	MB112T605	1	MB84049B	1
MB74LS145	2	MB14415	1	MB8516	1
MB74LS158	2	MB15021	1	MB88401 (709M)	1

Undocumented devices

Nine positions are custom or semi-custom Fujitsu silicon with no public datasheet: one is identified with confidence, one is probable, and **seven remain unknown**. The FM-7 sheets in this set draw the equivalent functions as discrete logic, so tracing which FM-7 circuit block disappeared into which gate array is the practical route to identifying them.

Pos.	Marking	Status
M29	MB15021	Identified: custom digital colour palette LSI (not a gate array)
M39	MB60H010A	Likely the display controller; 64-pin custom, no datasheet found
M62	MB14415	Unknown. On the FM-7 sheets MB14415 is the main-side main/sub interface gate array (PDF 7 / p.287) — start there
M38, M49, M55, M79, M80	MB112T601-605	Fujitsu semi-custom gate arrays; individual functions unknown
M72	MB111S601	Fujitsu semi-custom gate array; function unknown